Yaswanth Muppalla

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AI Platform Engineer | ***Senior Software Engineer*** | *Backend Systems* | ***AWS*** | *Data Engineering*

PROFESSIONAL EXPERIENCE

American Express | **Senior Software AI Engineer** **Apr 2023 - Present**

Impact: Built Java, **Kafka, Flink, Airflow**, and **API** services, IRIS-**backed RAG systems** for credit oversight workflows, improving reporting efficiency by **35%**, achieving **99% data accuracy**, reducing rerun/debug cycles by **3x**, and accelerating **AI copilot support**.

- Partnered with **BCA/CCA data pipeline teams** and built JSON-configurable **Apache Flink pipelines** for ingestion, transformation, joins, validation, and downstream publishing, translating workflow knowledge into **AI-ready pipeline patterns** for credit oversight use cases.
- Automated **Airflow orchestration and API** facing downstream publishing for credit data processing with dependency management, rerun handling, **SLA monitoring**, data-quality, and failure alerting, creating reliable signals for downstream **AI copilot and RAG workflows**.
- Designed an **IRIS-backed ingestion pipeline** with document extraction, deduplication, upsert handling, segmentation, **chunking**, metadata enrichment, and batch publishing for **semantic retrieval**.
- Developed a **RAG agent harness** that combined IRIS semantic search, scoped retrieval, prompt assembly, **source-ledger tracking**, and **grounded response generation** for credit oversight users.
- Built a **context harness** for chatbot-style interactions, managing **session context**, selected documents, retrieved chunks, policy context, user query intent, and **controlled LLM input assembly**.
- Added **observability and control layers** across the AI workflow, including retrieval logs, execution traces, citation mapping, **guardrails**, cache behavior, and **audit-friendly response tracking**.

Skechers | Software Engineer **Oct 2020 - Mar 2023**

Impact: Delivered **15+ Python APIs** and event-driven services, maintained **99.9% uptime**, reduced costs by **22%**, migrated **35M+ records**, and cut incident resolution time by **40%**.

- Developed **15+ Python/Pydantic REST APIs** on microservices-style, event-driven **AWS Lambda/Step Functions**, sustaining **99.9% uptime** and cutting costs **22%**.
- Migrated **35M+ SFMC records** into **MongoDB** and improved query performance by **50%** through indexing, caching, and data-access optimization.
- Integrated **SessionM**, SFMC, MongoDB, and **AWS serverless services** for loyalty/marketing workflows and **sub-500ms transactions**.
- Applied **event-driven API architecture** across Lambda, **Step Functions**, validation, orchestration, and data-access layers.
- Improved **production resilience** with CloudWatch/Datadog alerts, **API failure tracing**, **retry handling**, and SLA-focused incident feedback.

SOCH Inc. | Full-Stack Developer - Richmond, VA **Jan 2020 - Sep 2020**

- Built a **Spring Boot** ticketing system with **REST microservices** and full-stack workflow screens for government clients.
- Used **Java ExecutorService/CompletableFuture** for async ticket updates, notifications, and background job handling.
- Managed **concurrency** with synchronized services, **thread-safe collections**, timeouts, and Spring Boot exception handling.

FOUNDER-LED TECHNICAL PROJECTS

OmInsights - **Founder-Led AI Platform Build** **Jan 2024 - Present**

Impact: Built a **6-component AI platform** with **tenant-scoped API's**, Context Engine orchestration, MCP tool execution, async file processing, and **source-backed answers** over owned documents.

- Architected **Angular UI**, **FastAPI Business Service**, **Context Engine**, **MCP Server**, File Processing worker, and AWS-backed deployment paths with CI/CD and tenant-aware AWS operations.
- Built **document intelligence pipelines** for upload, parse, normalize, chunk, embed, persist, retrieve, and cite flows for **grounded RAG responses** with **source ledgers** and retrieval quality controls.
- Designed **Context Engine flows** for **chat/session memory**, selected context, KB injection, prompt assembly, and **LLM/context harness design** for multi-turn RAG orchestration and scoped answer grounding.
- Implemented **MCP/API runtime patterns** with **tool catalog APIs**, native/template tools, **scoped document search**, execution logging, cache behavior, and access-aware tool selection.
- Combined **source-ledger**, **citation**, guardrail, caching, and **semantic caching** patterns to keep AI outputs grounded and traceable.

TECHNICAL SKILLS

AI/LLM Systems: RAG, embeddings, **vector search**, LangChain/LangGraph, LlamaIndex/LlamaParse, **MCP/FastMCP**, OpenAI APIs, **Amazon Bedrock**, prompt orchestration, agent workflows/evaluation, LLMOps, semantic caching, context orchestration

Backend: Python, **FastAPI**, Pydantic, **Java 17**, **Spring Boot**, **REST APIs**, JWT, microservices, Java concurrency, TypeScript/Angular

Cloud & Data: AWS Lambda, ECS/S3/SQS, **Step Functions**, API Gateway/EventBridge, Docker, CI/CD, MongoDB, Redis, MemSQL, Kafka, **Flink/Airflow**

Retrieval & Storage: OpenSearch Vector DB, MongoDB, hybrid search, metadata filtering/reranking, chunking, **source ledgers**, searchable publishing, Elasticsearch

Observability: Datadog, **CloudWatch**, Kibana, X-Ray, ELK Stack, **structured logging**, SLA alerts, failure tracing

EDUCATION

State University of New York, New Paltz | Master of Science, Computer Science | CGPA: 3.6/4.0 May 2020

JNTUK | Bachelor of Technology | CGPA: 8.5/10 2018